LAPORAN
WORKSHOP DEVELOPER OPERATIONAL
MINGGU KE-9

MATERI PEMBELAJARAN

Implementasi Docker Swarm

Oleh

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PROGRAM STUDI TEKNIK INFORMATIKA

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

2026

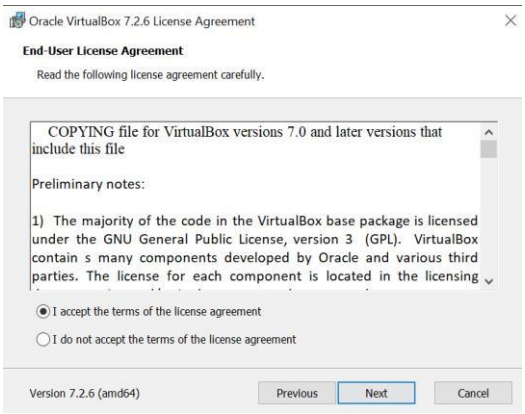
HASIL DAN PEMBAHASAN

A. Instalasi Virtual Box

1. Lakukan instalasi menggunakan Virtual Box melalu link berikut Downloads – Oracle VirtualBox
2. Selesaikan proses download hingga muncul tampilan di bawah ini dan klik Next.

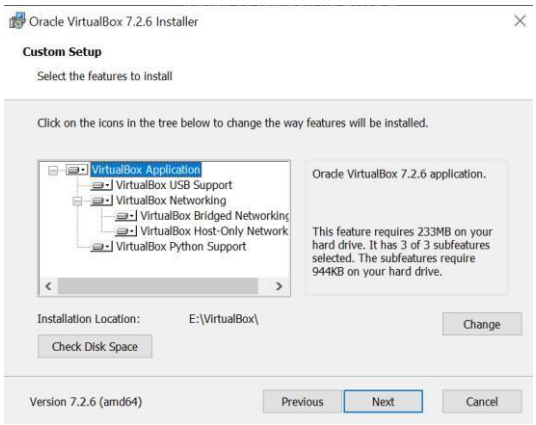
Gambar 1. 1 Tampilan Install Virtual Box

3. Pilih I Accept… dan akan muncul button Next dan klik Next.



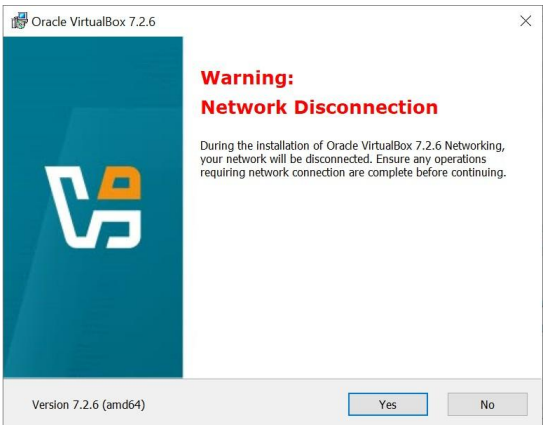
Gambar 1. 2 License Agrement

4. Atur dimana file hasil instalasi akan disimpan. Pastikan ruangan yang dimiliki melebihi 10gb ya.



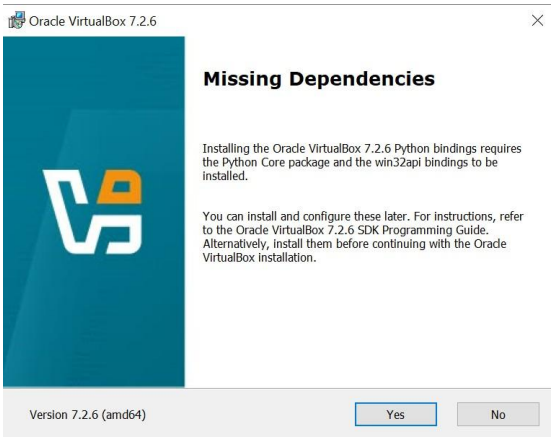
Gambar 1. 3 Setup Installtion Location

5. Jangan panik jika muncul tampilan seperti di bawah. Ini hanya memastikan koneksi internet yang kamu gunakan stabil ya. Kemudian klik saja Next.



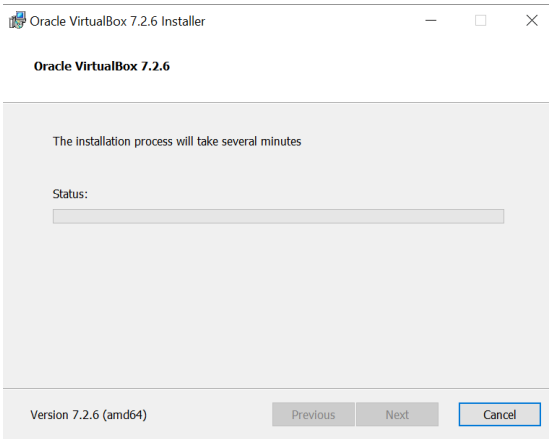
Gambar 1. 4 Warning Network

6. Setelah itu, klik yes lagi pada tampilan seperti di bawah ini. Hal ini untuk menunjukan informasi pada Virtual Box.



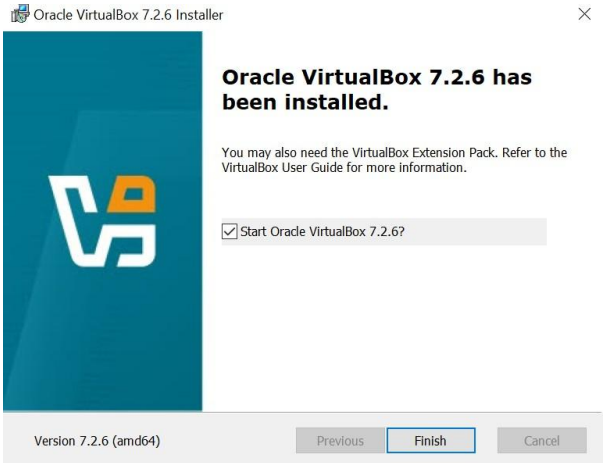
Gambar 1. 5 Missing Dependencies

7. Tunggu proses instalasi hingga selesai ya. Harap bersabar dan selalu menunggu.



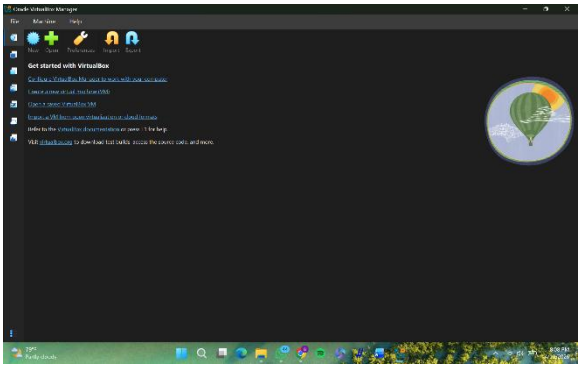
Gambar 1. 6 Proses Installation

8. Proses instalasi selesai ditandai dengan tampilan seperti berikut. Kemudian klik finish.



Gambar 1. 7 Instalasi Selesai

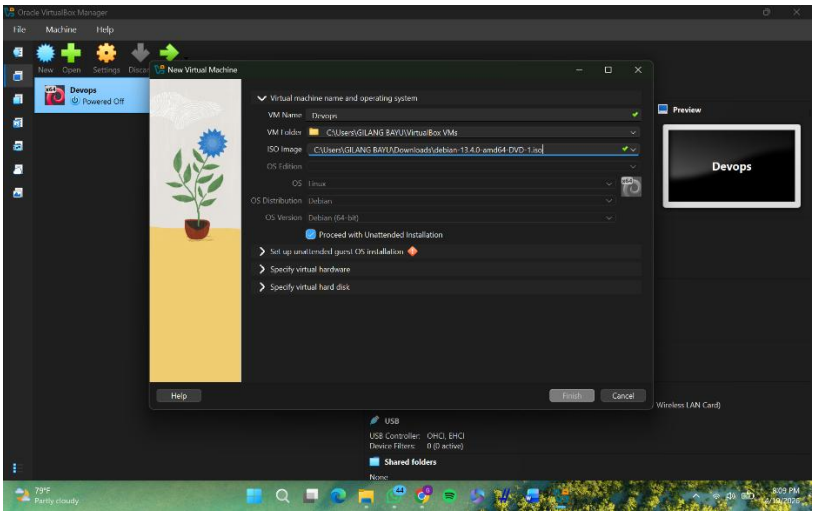
9. Tampilan saat membuka Virtual Box



Gambar 1. 8 Tampilan Utama Virtual Box

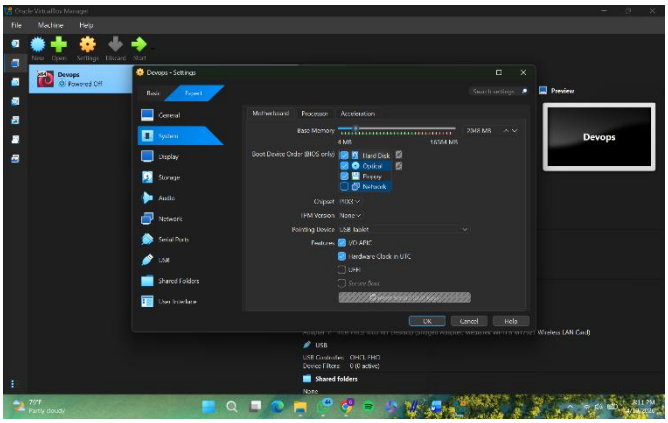
B. Pembuatan Debian

1. Memberikan nama Vm sesuaikan dengan kebutuhan, misalnya seperti di bawah ini.



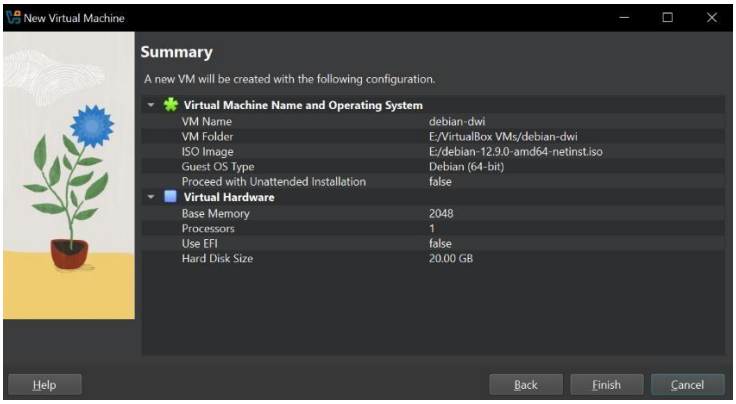
Gambar 1. 9 New Virtual Machine

2. Setting hardware yang akan digunakan.



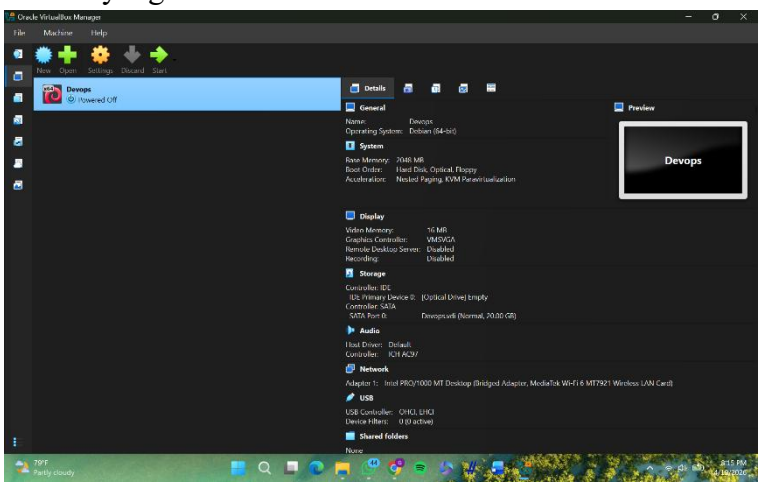
Gambar 1. 10 Specify Virtual Hardware

3. Tampilan spesifikasi VM yang sudah kita buat



Gambar 1. 11 Summary

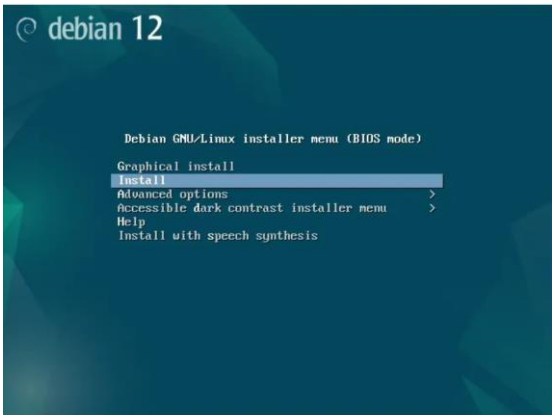
4. Tampilan debian yang sudah dibuat.



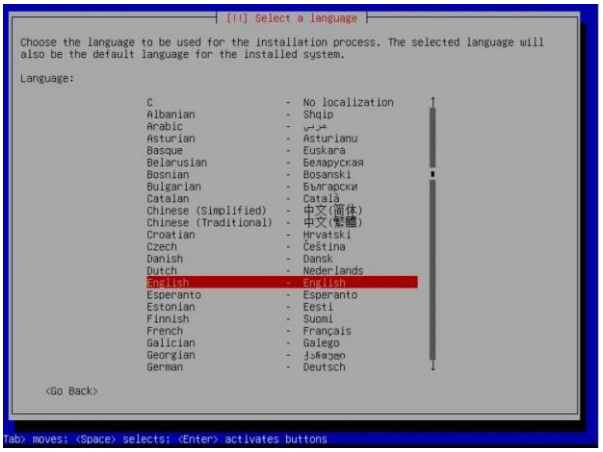
Gambar 1. 12 Debian

C. Instalasi Debian

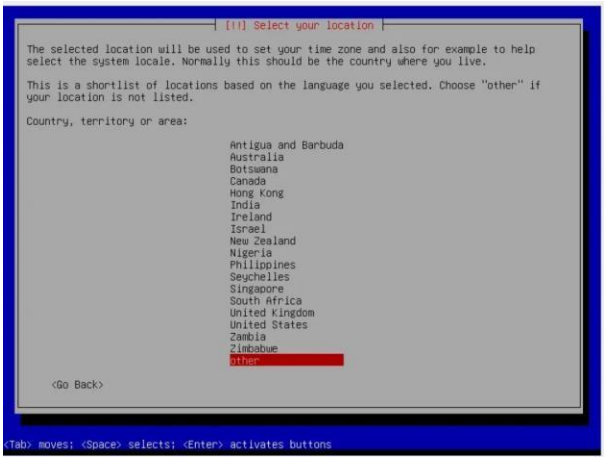
Lakukan proses Instalasi debian seperti tahapan di gambar bawah ini



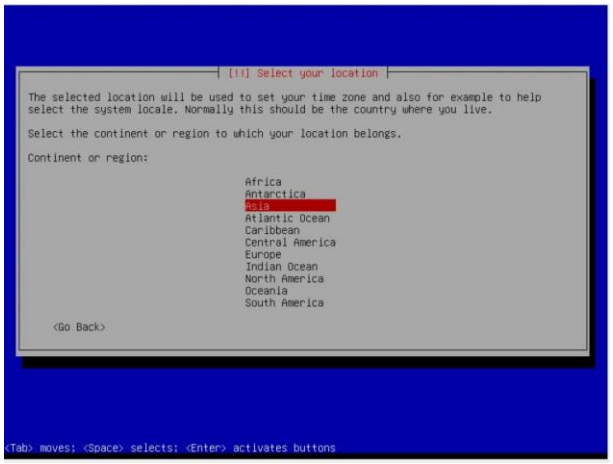
Gambar 1. 13 Instalasi Debian



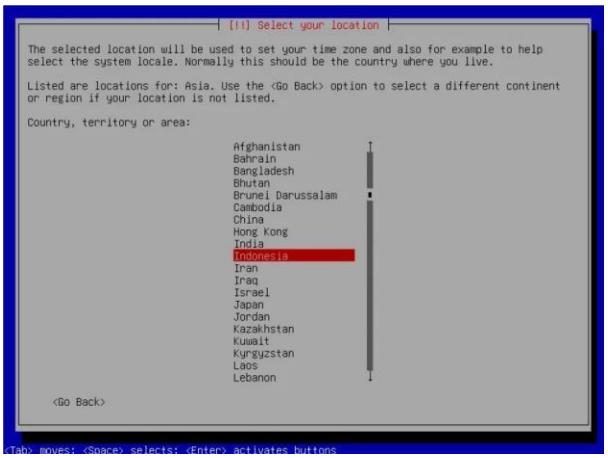
Gambar 1. 14 Language Install



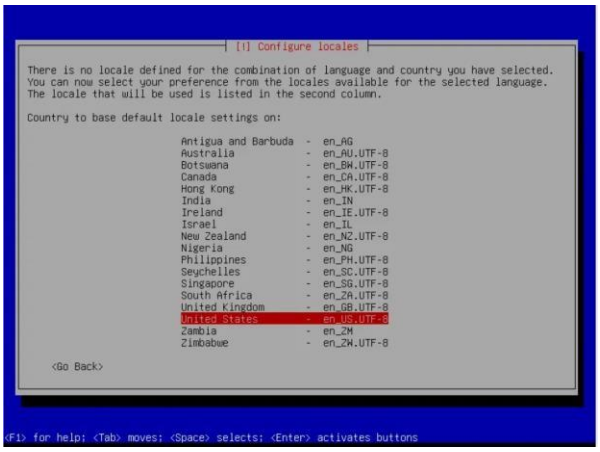
Gambar 1. 15 Setting Location



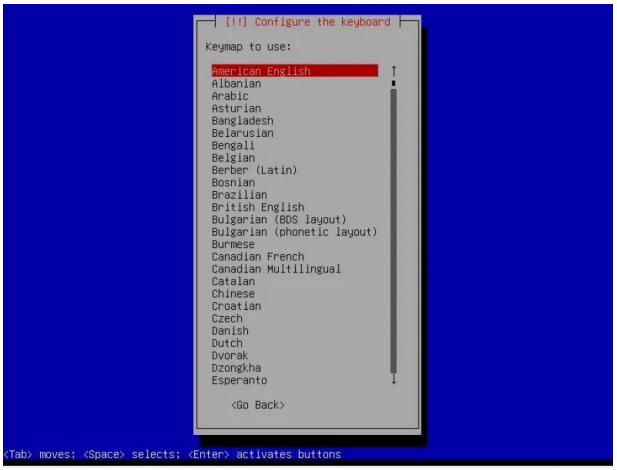
Gambar 1. 16 Region



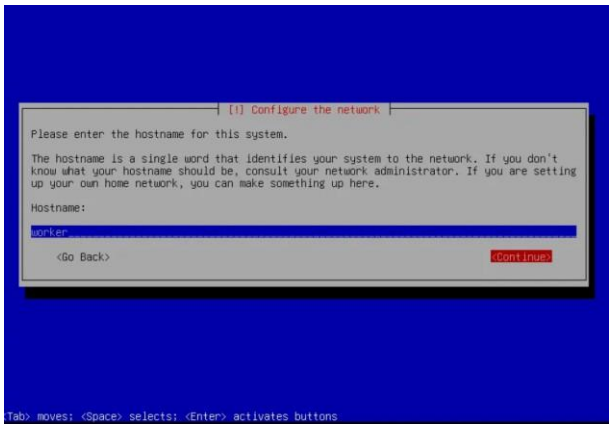
Gambar 1. 17 Location



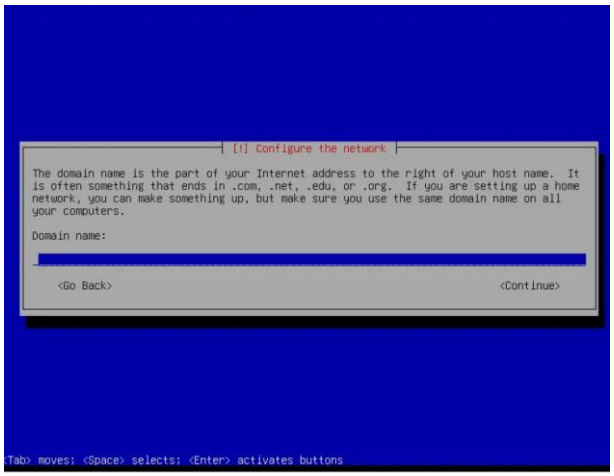
Gambar 1. 18 Locales



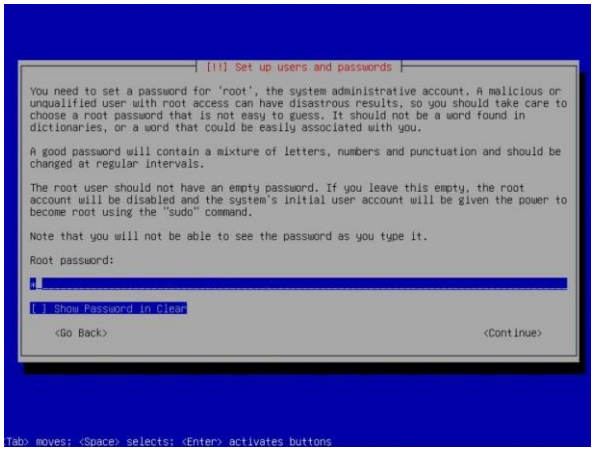
Gambar 1. 19 Setting Keyboard



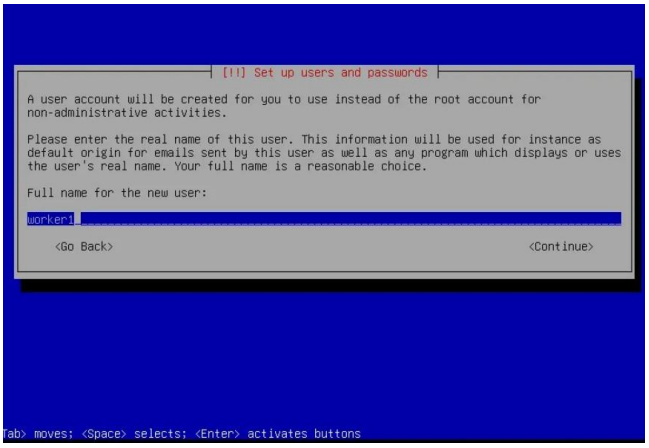
Gambar 1. 20 Configure Network



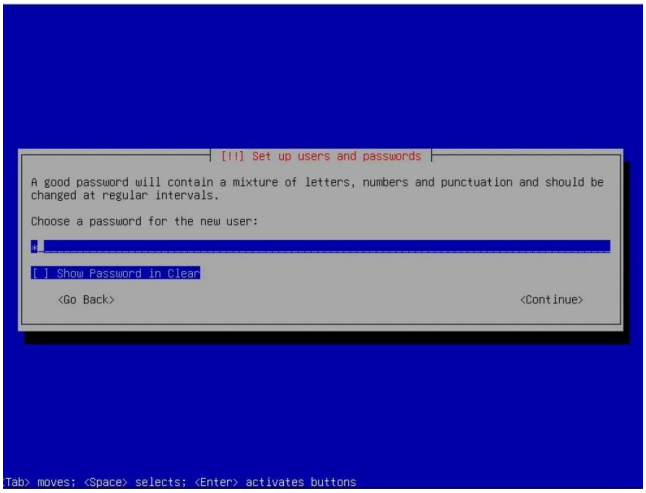
Gambar 1. 21 Domain



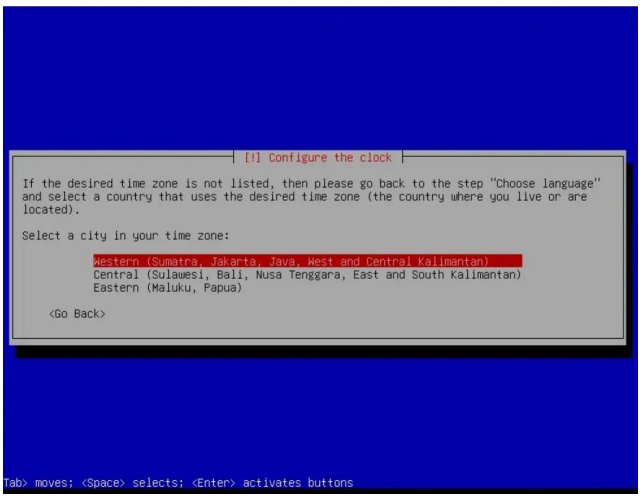
Gambar 1. 22 Password



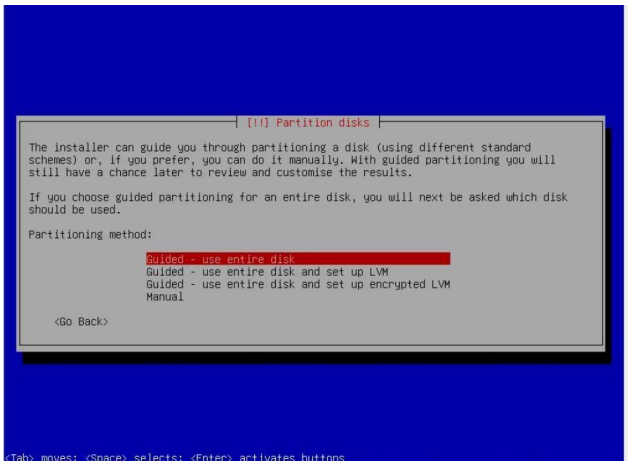
Gambar 1. 23 Name User



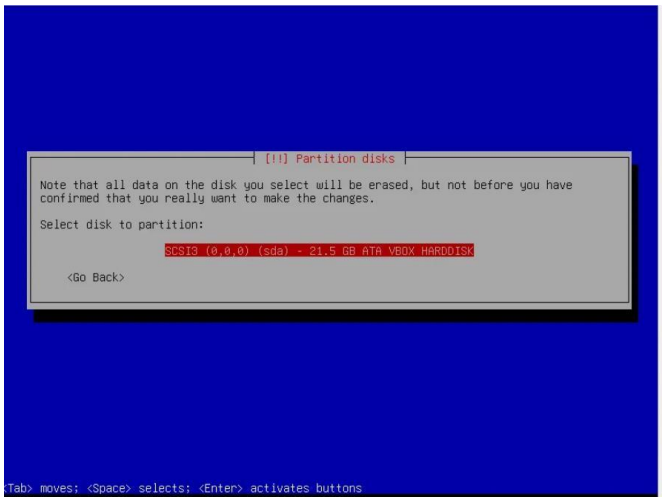
Gambar 1. 24 Password



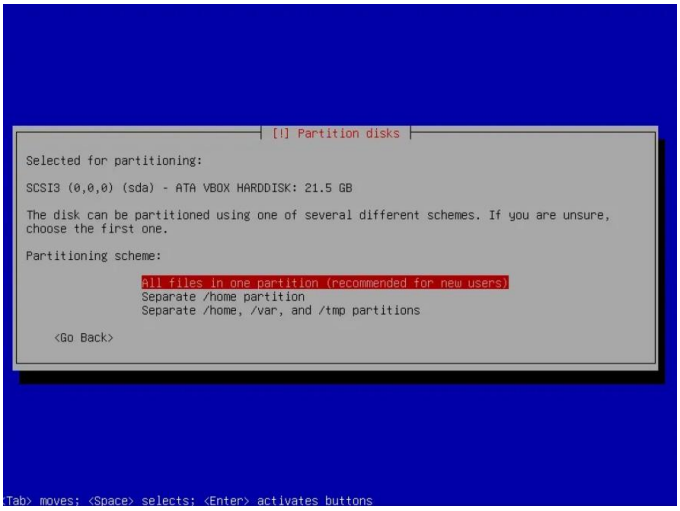
Gambar 1. 25 Clock



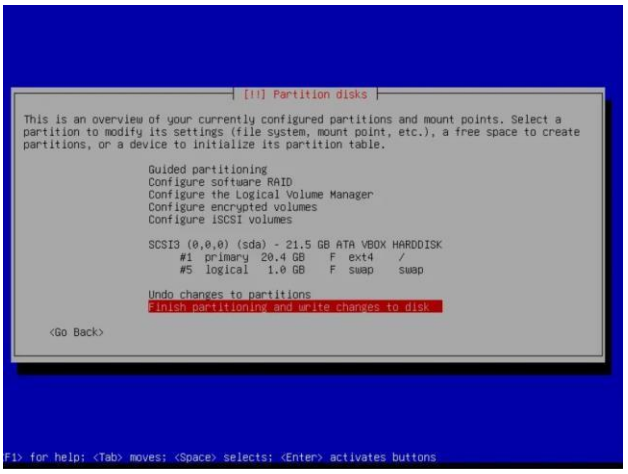
Gambar 1. 26 Partisian Disk



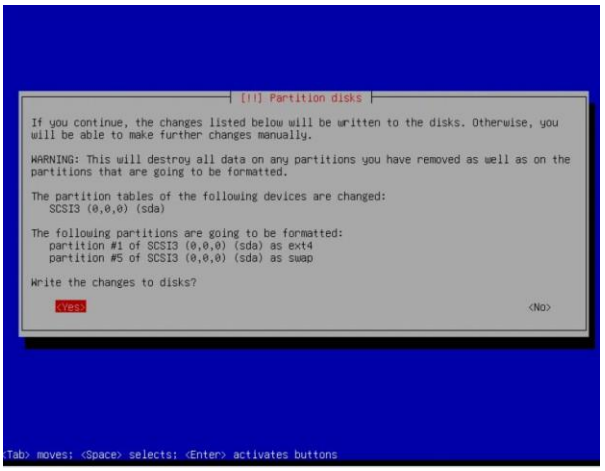
Gambar 1. 27 Select Disk



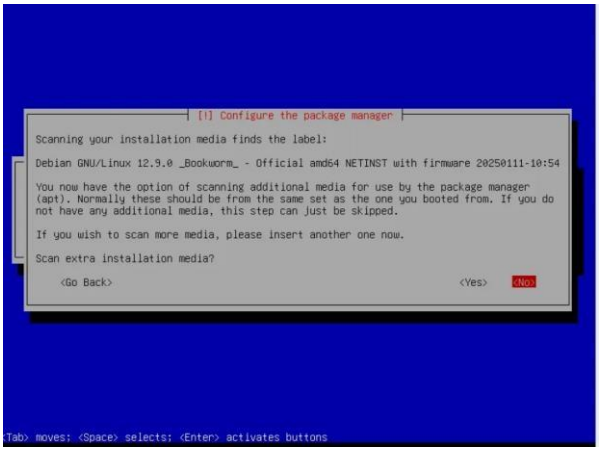
Gambar 1. 28 Scheme



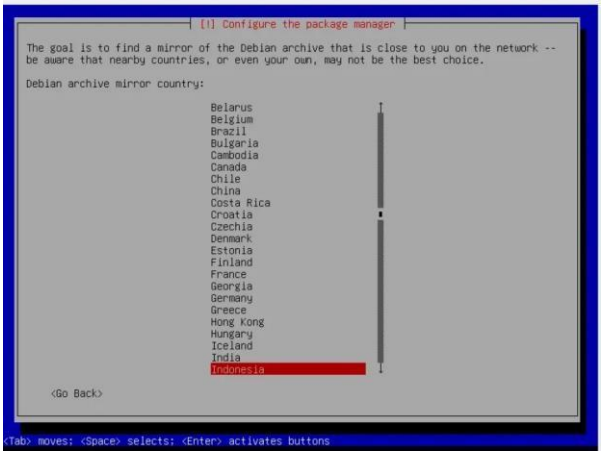
Gambar 1. 29 Finish Change Disk



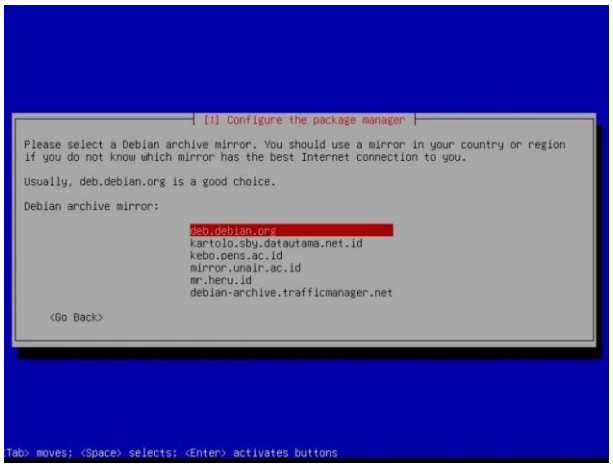
Gambar 1. 30 Write



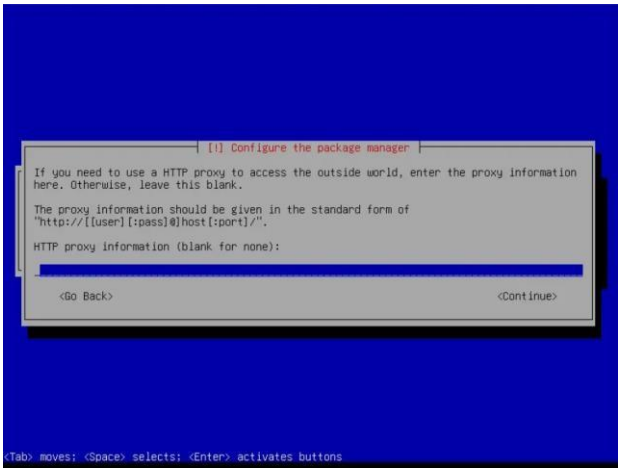
Gambar 1. 31 Scan extra



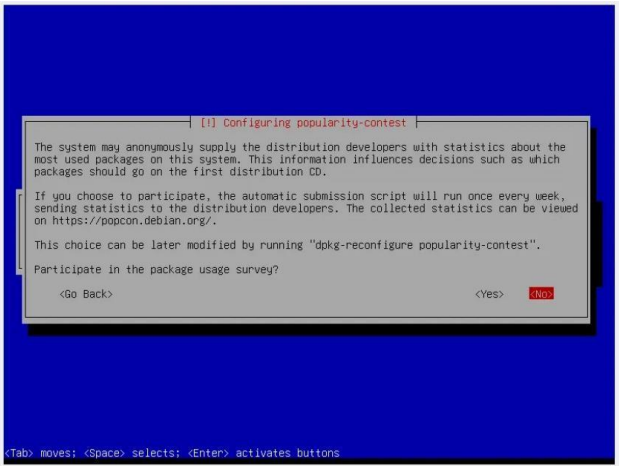
Gambar 1. 32 Mirror Country



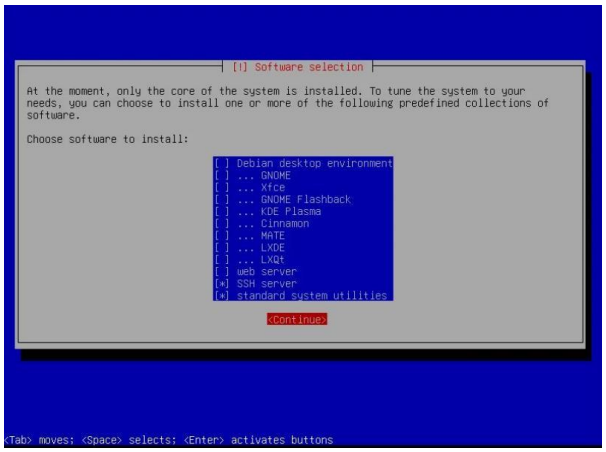
Gambar 1. 33 Debian Archive



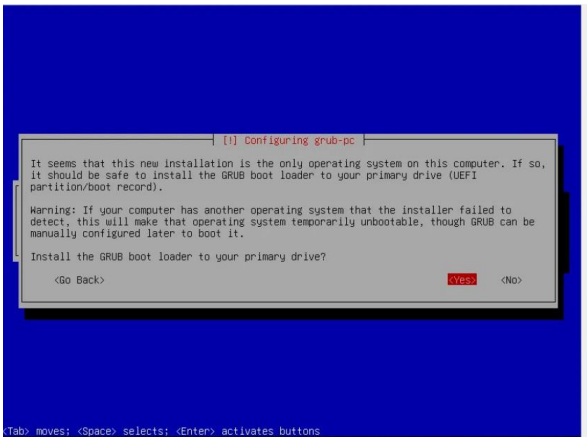
Gambar 1. 34 HTTPProxy



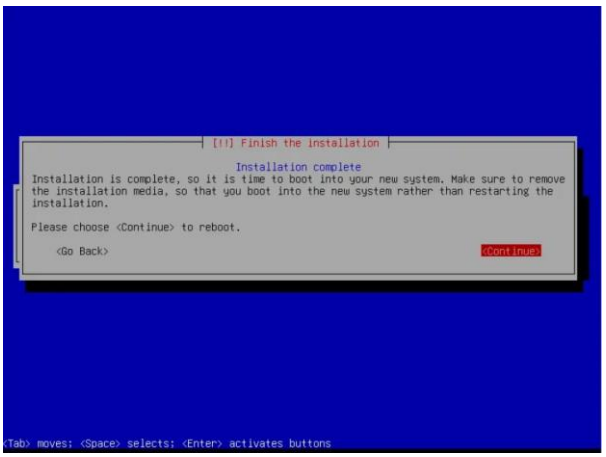
Gambar 1. 35 Participate



Gambar 1. 36 Chose Software



Gambar 1. 37 Install GRUB



Gambar 1. 38 Installation Complete

D. Implementasi Docker Swarm

1. Saya disini menggunakan 2 os, dimana disini debian pertama sebagai manager dan wsl ubuntu kedua sebagai worker.

2. Sebelum menghubungkan dua node, hal pertama yang dipastikan adalah konektivitas antar perangkat.

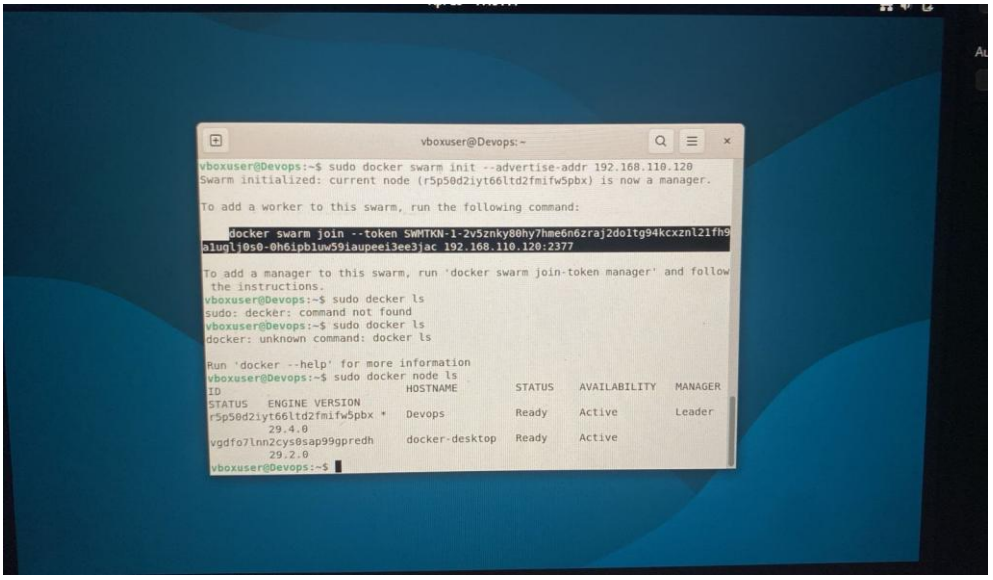
    - **Identifikasi IP Address:** Menggunakan perintah *ip a* pada node Manager untuk mengetahui IP yang aktif (misal: 192.168.110.120).

    - **Pengujian Koneksi:** Melakukan ping dari WSL Ubuntu ke IP Manager untuk memastikan tidak ada hambatan komunikasi di jaringan lokal.

    - **Konfigurasi Firewall:** Memastikan port-port standar Docker Swarm (2377, 7946, 4789) tidak diblokir oleh firewall (ufw di Linux atau Firewall di Windows Host).

3. Inisialisasi Swarm Manager (Node Utama)

    - Dengan Perintah *sudo docker swarm init --advertise-addr 192.168.110.120*

    - **Output:** Sistem menghasilkan sebuah **Join Token** unik yang akan digunakan oleh node Worker untuk bergabung.

4. Terus copy paste token yang di dapat pada terminal wsl ubntu, setelah berhasil terhubung kita bisa melihat hasil nya dengan melakukan perintah pada node yang menjadi manager dengan malakukan ***sudo docker node ls.***